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Ex.No. 11: Implementation of IoT-based Smart Agriculture using MATLAB

AIM: To design and simulate an IoT-based smart agriculture system using MATLAB by monitoring soil moisture levels, controlling water pump status (ON/OFF), and evaluating irrigation efficiency for optimized water resource management.

Objective:

1. To monitor the Soil Moisture (%) level in agricultural fields using IoT-based sensing simulation in MATLAB.
2. To control and display the Water Pump Status (ON/OFF) based on threshold soil moisture conditions.
3. To evaluate the Irrigation Efficiency (%) by analyzing water usage effectiveness and moisture maintenance level.

Procedure:

```
clc; clear; close all;
```

```
% Simulated soil moisture data (%)
```

```
time = 1:10;
```

```
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
```

```
% Threshold for irrigation
```

```
threshold = 40;
```

```
figure
```

```
% ----- Graph 1: Soil Moisture Variation -----
```

```
subplot(2,1,1)
```

```
plot(time, soil_moisture, '-o', 'LineWidth', 2)
```

```
title('Soil Moisture Monitoring in IoT Smart Agriculture')
```

```
xlabel('Time (Hours)')
```

```
ylabel('Soil Moisture (%)')
```

```
grid on
```

```
hold on
```

```
ylines(threshold,'r--','Threshold')
```

```
% ----- Graph 2: Moisture Status (Dry/Wet Level) -----
```

```
status = soil_moisture >= threshold;
```

```
subplot(2,1,2)
```

```
stem(time, status, 'filled', 'LineWidth', 2)
```

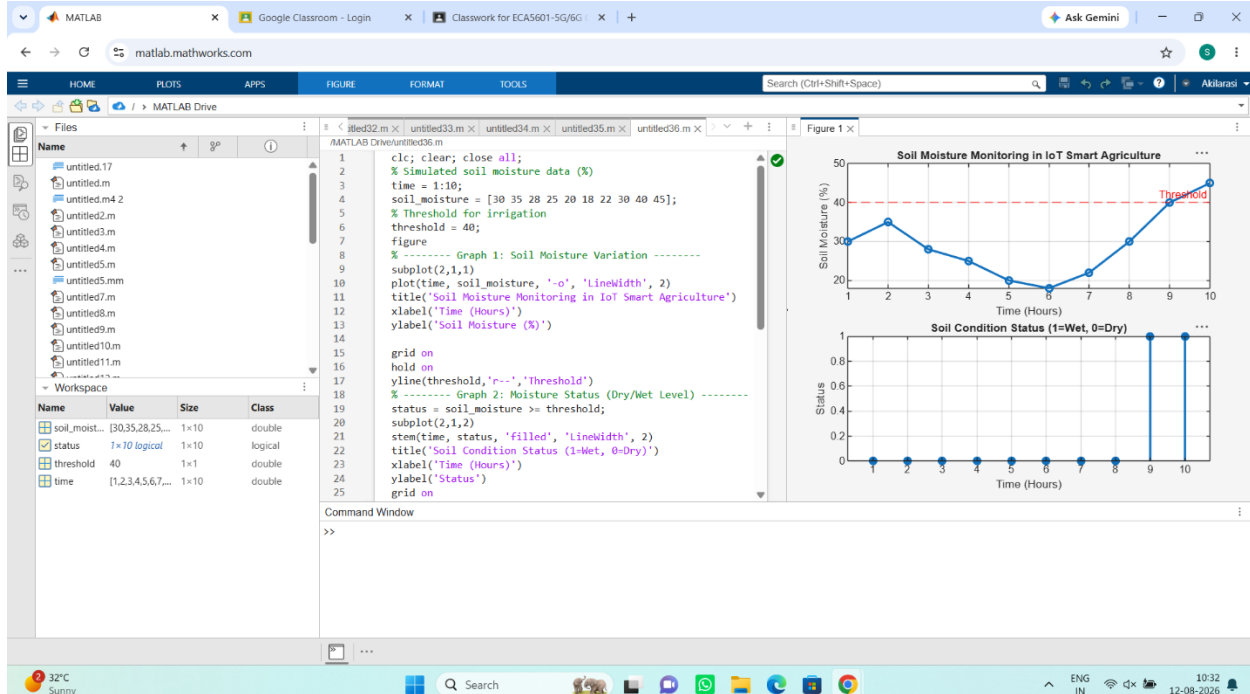
```
title('Soil Condition Status (1=Wet, 0=Dry)')
```

```
xlabel('Time (Hours)')
```

```
ylabel('Status')
```

```
grid on
```

Output:



Result: Soil moisture levels vary over time, showing fluctuations in field water content. The threshold line helps identify dry and wet conditions for smart irrigation decisions.

Objective 2:

```
clc; clear; close all;

% Simulated soil moisture (%)

time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level

threshold = 40; % Pump control logic (1 = ON, 0 = OFF)

pump_status = soil_moisture < threshold;

figure

% ----- Graph 1: Soil Moisture vs Pump Status -----

subplot(2,1,1)

plot(time, soil_moisture, '-o','LineWidth',2)

title('Soil Moisture vs Pump Control')

xlabel('Time (Hours)')

ylabel('Soil Moisture (%)')

grid on

hold on

yline(threshold,'r--','Threshold')

% ----- Graph 2: Pump Status (ON/OFF) -----

subplot(2,1,2)

stem(time, pump_status, 'filled','LineWidth',2)

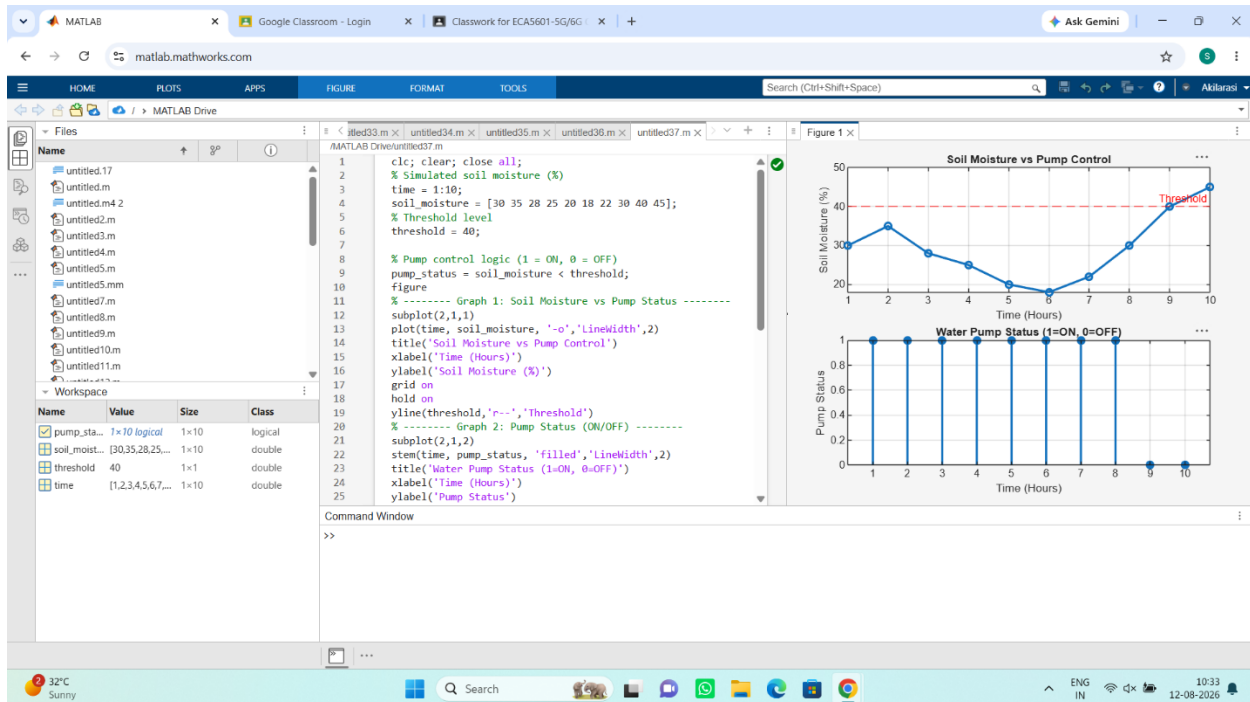
title('Water Pump Status (1=ON, 0=OFF)')

xlabel('Time (Hours)')

ylabel('Pump Status')

grid on
```

Output:



Result: The pump turns ON automatically when soil moisture drops below the threshold level. When moisture is sufficient, the pump remains OFF, ensuring efficient water usage and smart irrigation control.

Objective 3:

clc; clear; close all;

% Simulated soil moisture data (%)

time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level

threshold = 40; % Irrigation efficiency calculation (%)

efficiency = (soil_moisture ./ threshold) * 100;

efficiency(efficiency > 100) = 100;

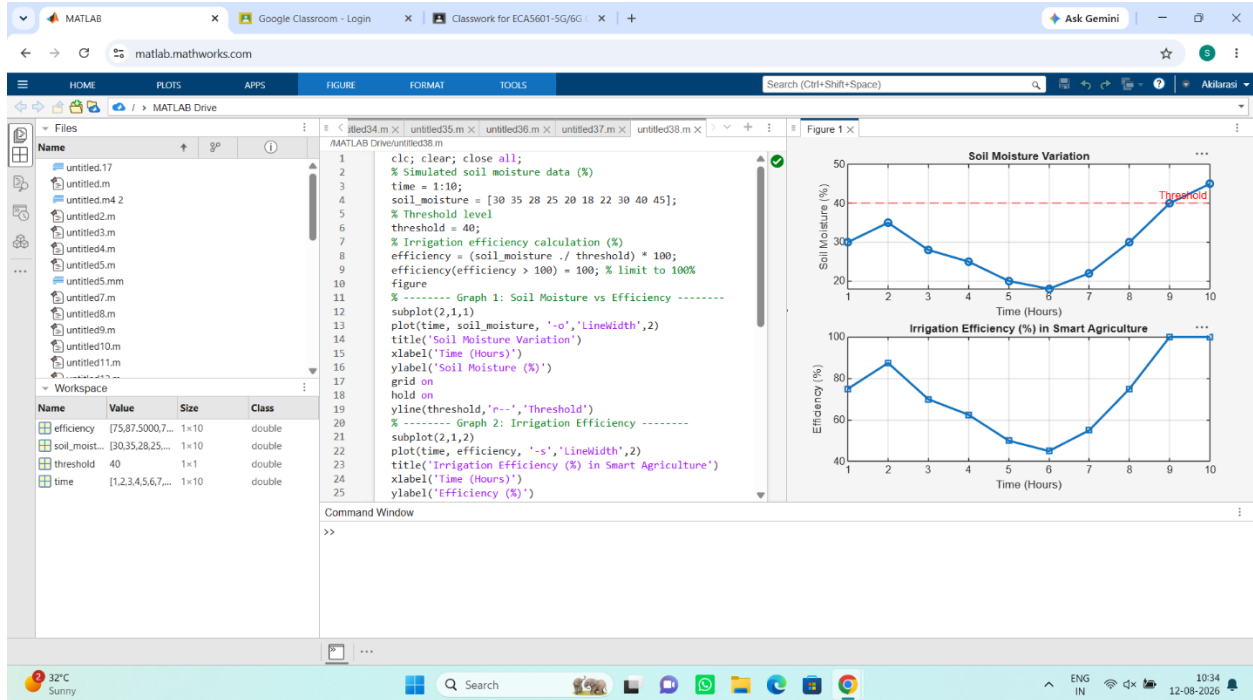
% limit to 100%

figure

% ----- Graph 1: Soil Moisture vs Efficiency -----

```
subplot(2,1,1)
plot(time, soil_moisture, '-o',LineWidth,2)
title('Soil Moisture Variation')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold,'r--','Threshold')
% ----- Graph 2: Irrigation Efficiency -----
subplot(2,1,2)
plot(time, efficiency, '-s',LineWidth,2)
title('Irrigation Efficiency (%) in Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Efficiency (%)')
grid on
```

Output:



Result: Irrigation efficiency increases when soil moisture approaches the optimal threshold level. The system ensures better water utilization by maintaining soil moisture within the desired range using smart control.